

PRODUCT REQUIREMENTS DOCUMENT

Fuel Station Management System

Odoo 15 — Complete End-to-End Module

Module Name	fuel_station
Framework	Odoo 15 (Community Edition)
Developer	Anushka
Type	Complete ERP Module — built manually
Covers	Master data, Inventory, Shifts, Sales, Accounting, Fleet, Website

1. Business Overview

A fuel station management system manages the complete lifecycle of a petrol station — from fuel procurement and tank storage, through pump and nozzle management, employee shifts, meter readings, fuel sales, customer accounts, fleet management, and full accounting integration. This PRD covers the complete Odoo 15 implementation built manually from scratch.

Core Business Flow

Fuel Types	Define petrol, diesel, premium fuel types
Tanks	Underground storage tanks — each holds one fuel type
Pumps	Each pump connects to one tank
Nozzles	Each pump has nozzles with meters counting litres dispensed
Shifts	Employee opens shift, records opening meter readings
Sales	Customer fills fuel — meter difference = litres sold
Purchase	Manager orders fuel when tank runs low — stock updated on delivery
Accounting	Sales reconciled against cash — expenses tracked — P&L generated
Website	Public pages — fuel prices, branch locator, customer portal

2. Technology Stack

Layer	Technology	Purpose
Backend	Python 3.8+, Odoo ORM	Models, fields, business logic, workflow methods
Views	XML	Form, List, Search, Kanban views
Reports	QWeb + wkhtmltopdf	PDF reports for shifts, sales, tanks
Excel	xlsxwriter / openpyxl	Excel export for all reports
Styling	CSS / SCSS	Custom UI styling via assets_backend
Frontend	JavaScript (Odoo 15 legacy)	Custom widgets where needed
Accounting	Odoo account module	Journal entries, invoices, payments
Website	Odoo website module + QWeb	Public pages and customer portal
Database	PostgreSQL 14	Managed via Docker
Environment	Docker + docker-compose	Odoo 15 + PostgreSQL containers

3. Module File Structure

```
fuel_station/  
■■■ __init__.py
```

```

■■■■ __manifest__.py
■■■■ models/
■ ■■■■ __init__.py
■ ■■■■ fuel_type.py
■ ■■■■ fuel_tank.py
■ ■■■■ fuel_pump.py
■ ■■■■ fuel_nozzle.py
■ ■■■■ fuel_shift.py
■ ■■■■ fuel_sale.py
■ ■■■■ fuel_purchase.py
■ ■■■■ fuel_expense.py
■ ■■■■ fuel_fleet.py
■■■■ views/
■ ■■■■ fuel_type_views.xml
■ ■■■■ fuel_tank_views.xml
■ ■■■■ fuel_pump_views.xml
■ ■■■■ fuel_nozzle_views.xml
■ ■■■■ fuel_shift_views.xml
■ ■■■■ fuel_sale_views.xml
■ ■■■■ fuel_purchase_views.xml
■ ■■■■ menu.xml
■■■■ report/
■ ■■■■ shift_report.xml
■ ■■■■ daily_sales_report.xml
■ ■■■■ tank_level_report.xml
■■■■ security/
■ ■■■■ ir.model.access.csv
■■■■ website/
■ ■■■■ templates/
■ ■■■■ home.xml
■ ■■■■ about.xml
■ ■■■■ fuel_prices.xml
■ ■■■■ contact.xml
■■■■ static/
■ ■■■■ src/
■ ■■■■ css/
■ ■■■■ fuel_station.css

```

4. Data Models

Model (Technical Name)	Key Fields
fuel.type	name, code, price_per_litre, active
fuel.tank	name, fuel_type_id, capacity, current_stock, min_level, location
fuel.pump	name, tank_id, location, active
fuel.nozzle	name, pump_id, fuel_type_id, current_meter, active
fuel.shift	name, employee_id, date, start_time, end_time, state (open/closed), opening_readings, closing_readings
fuel.sale	shift_id, nozzle_id, litres_sold, unit_price, total_amount, payment_method, customer_id
fuel.purchase	supplier_id, fuel_type_id, tank_id, quantity, unit_price, total_cost, delivery_date, state
fuel.expense	name, shift_id, amount, expense_type, description
fuel.fleet	vehicle_number, customer_id, fuel_type_id, credit_limit, balance

5. Implementation Plan — Step by Step

Phase 1 — Module Scaffold

Step 1

Create fuel_station/ folder in custom_addons

The root folder Odoo scans for modules

Step
2

Create `__init__.py` (root)

Marks folder as Python package

Step
3

Create `__manifest__.py`

Registers module with Odoo — name, version, depends, data files

Phase 2 — Master Data Models

Step
4

`models/ __init__.py`

Imports all model files so Odoo can discover them

Step
5

`fuel_type.py` — `fuel.type`

Define fuel types: petrol, diesel, premium

Step
6

`fuel_tank.py` — `fuel.tank`

Underground tanks with capacity and stock level

Step
7

`fuel_pump.py` — `fuel.pump`

Pumps connected to tanks

Step
8

`fuel_nozzle.py` — `fuel.nozzle`

Nozzles on pumps with meter readings

Phase 3 — Operations Models

Step
9

`fuel_shift.py` — `fuel.shift`

Shift management — open/close with meter readings

Step
10

`fuel_sale.py` — `fuel.sale`

Sales calculated from meter differences per shift

Step
11

`fuel_purchase.py` — `fuel.purchase`

Fuel procurement from suppliers — updates tank stock

Step
12

`fuel_expense.py` — `fuel.expense`

Expenses per shift — cash, salary, electricity

Step
13

`fuel_fleet.py` — `fuel.fleet`

Fleet vehicles with credit accounts

Phase 4 — Security

Step
14

`ir.model.access.csv`

Access rules for every model — mandatory before install

Phase 5 — XML Views

Step
15

`fuel_type_views.xml`

Form, List, Search views for fuel types

Step
16

`fuel_tank_views.xml`

Tank management views with stock indicator

Step 17	fuel_pump_views.xml Pump views linked to tanks
Step 18	fuel_nozzle_views.xml Nozzle views with meter reading display
Step 19	fuel_shift_views.xml Shift form with opening/closing meter readings and status bar
Step 20	fuel_sale_views.xml Sales views with payment method and reconciliation
Step 21	fuel_purchase_views.xml Purchase order views with delivery workflow
Step 22	menu.xml All menus, sub-menus and window actions

Phase 6 — Reports

Step 23	shift_report.xml + template PDF summary of each shift — sales, cash, expenses
Step 24	daily_sales_report.xml + template Daily sales by fuel type and nozzle
Step 25	tank_level_report.xml + template Current stock level of all tanks
Step 26	Excel export — sales, shifts, purchases xlsxwriter-based Excel reports for all modules

Phase 7 — Accounting Integration

Step 27	Link fuel.sale to account.move Auto-generate journal entries on each sale
Step 28	Link fuel.purchase to vendor bills Vendor bill created on fuel delivery
Step 29	Shift reconciliation Cash expected vs cash received per shift
Step 30	Dashboard — sales, stock, revenue Odoo dashboard with key metrics

Phase 8 — Website

Step 31	Home page Station overview, current fuel prices
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Step 32	About Us page Company info and branch details
Step 33	Fuel Prices page Live prices pulled from fuel.type model
Step 34	Branch Locator page Map with branch locations
Step 35	Fleet Registration page Online form for fleet vehicle registration
Step 36	Contact Us page Contact form with email integration
Step 37	Customer Login Portal Customer can view their account, credit, and fuel history

6. Reports & Exports

Report Name	Format	Description
Shift Summary Report	PDF	Sales, cash, expenses per shift
Daily Sales Report	PDF	Revenue breakdown by fuel type
Tank Level Report	PDF	Current stock across all tanks
Purchase History Report	PDF	Supplier orders and deliveries
Customer Statement	PDF	Fleet customer credit and usage
Sales Export	Excel	Full sales data filterable by date/shift
Shift Export	Excel	All shifts with reconciliation data
Purchase Export	Excel	All procurement records
Tank Level Export	Excel	Historical stock levels

7. Odoo Website Pages

Home	Station overview, live fuel prices, quick links
About Us	Company history, mission, team, branch info
Fuel Prices	Live prices pulled from fuel.type model in Odoo
Branch Locator	Map showing all station locations
Fleet Registration	Online form — vehicle registration for credit accounts
Contact Us	Contact form with email notification
Customer Login	Portal — view account balance, credit limit, fuel history
Services	List of station services (car wash, oil change, etc.)

8. Development Timeline

Week 1	Module scaffold + all master data models (Steps 1–8)
Week 2	Operations models — shift, sale, purchase, fleet (Steps 9–13)
Week 3	Security + all XML views + menus (Steps 14–22)
Week 4	PDF and Excel reports (Steps 23–26)
Week 5	Accounting integration + dashboard (Steps 27–30)

Week 6	Odoo website — all pages (Steps 31–37)
Week 7	Testing, bug fixes, demo to mentor

9. Success Criteria

- Module installs on Odoo 15 without any errors
- All models create correct database tables in PostgreSQL
- Complete shift workflow — open, record meters, close, reconcile
- Sales calculated automatically from meter readings
- Tank stock updates correctly on purchase delivery
- All PDF reports generate without errors
- All Excel exports download with correct data
- Odoo website pages are publicly accessible
- Customer portal login works with correct data
- Accounting entries created automatically on sales and purchases
- All code written manually — no AI-generated boilerplate
- Mentor approves each phase before moving to the next